CROSS-DEGREE UNIVERSALITY OF THE BOREL-SINGULARITY RADIUS FOR POLYNOMIAL CONTINUED FRACTIONS

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Abstract. We isolate, as a single citable artefact, the cross-degree identity

$$\xi_0(b) \;=\; \frac{d}{\beta_d^{1/d}}$$

for the leading Borel-plane singular distance of the formal generating function $f(z) = \sum_{n\geq 0} Q_n z^n$ of a polynomial continued fraction $(1, b)$ of degree d with leading coefficient $\beta_d > 0$. Version 2.1 (2026-05-03) closes the Borel-summability citation residual admitted in v2's annotated bibliography by anchoring the Borel-summability layer in Birkhoff–Trjitzinsky 1933 §§4–6 ([2]) instead of in Birkhoff 1930 §§2–3, which contained only the formal-series existence half. The artefact thereby moves from a theorem-with-documented-residual to a theorem with a closed citation chain at all $d \geq 2$ uniformly. The contributions are: (i) an explicit Newton-polygon characteristic-polynomial Lemma (Theorem 4.1) with a 3-step mechanical proof replacing the v2 "by Phase A* symbolic derivation" black-box; (ii) the Phase A* extended sweep of [7] at $d \in \{2, 3, \ldots, 10\}$, 18/18 pass at dps = 50, as numerical verification of the Lemma; and (iii) the Borel-summability citation chain via Birkhoff–Trjitzinsky 1933 §§4–6 + Wasow 1965 §19 (uniform-in-q asymptotic existence) + Birkhoff 1930 §2 (uniform-in-n formal-series existence). The Newton-polygon slope-$1/d$ edge / Wasow shearing-exponent equivalence is recorded as a substantive vocabulary correspondence. The note remains purely consolidatory: no new numerical pipeline is run.

1. Setup

Throughout this note $(1, b)$ denotes a polynomial continued fraction with constant numerator $a_n \equiv 1$ and polynomial denominator

$$b(n) \;=\; \beta_d\, n^d + \beta_{d-1}\, n^{d-1} + \cdots + \beta_1\, n + \beta_0, \qquad \beta_d > 0, \tag{1}$$

with $d \geq 2$. Let Q_n denote the partial denominators of the PCF, satisfying the three-term recurrence $Q_n = b(n)\, Q_{n-1} + Q_{n-2}$ with $Q_{-1} = 0$, $Q_0 = 1$. The formal generating function

$$f(z) \;=\; \sum_{n\geq 0} Q_n\, z^n$$

satisfies the order-2 inhomogeneous linear ODE

$$L_d f \;:=\; \big(1 \,-\, z\, B_d(\theta + 1) \,-\, z^2\big)\, f \;=\; 1, \qquad \theta := z\, \partial_z, \tag{2}$$

with B_d the d-th degree polynomial encoding b on the shift-rescaled Euler operator θ. Explicitly, $B_d(s)$ is the polynomial obtained from $b(n)$ by the substitution $n \mapsto s$: the identity $b(n) = B_d(n)$

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holds at every integer $n \geq 0$, and B_d acts on the generating function via $B_d(\theta+1)\,f$ pulling back the shifted-recurrence factor $b(n+1)$ in the z^n coefficient through the Euler operator θ. For clarity we record three specialisations:

$$\begin{aligned} d=2: \quad B_2(s) &= \beta_2\, s^2 + \beta_1\, s + \beta_0, \\ d=3: \quad B_3(s) &= \beta_3\, s^3 + \beta_2\, s^2 + \beta_1\, s + \beta_0, \\ d=4: \quad B_4(s) &= \beta_4\, s^4 + \beta_3\, s^3 + \beta_2\, s^2 + \beta_1\, s + \beta_0. \end{aligned}$$

Equation (2) has a regular point at $z=\infty$ and an irregular singular point at $z=0$; the Borel-plane singular distance ξ_0 is the modulus of the leading characteristic root at $z=0$ along the slope-$1/d$ edge of the Newton polygon of the homogeneous part of (2). The scope of this note is fixed by $\beta_d > 0$ and $d \geq 2$; the $\beta_d < 0$ branch and the so-called "$d=2$ anomaly" Galois bin ([5, Remark 5.E][1]) are out of scope and are recorded as open in Section 5.

1.1. **PCF degree ↔ irregular-singularity rank.** Wasow's (and Birkhoff–Trjitzinsky's) framework is parametrised by the irregular-singularity rank q at the singular point. We derive the mapping $q=(d+2)/2$ from (2) as follows. Substitute $z=u^d$ on (2); then $\theta = z\,\partial_z = (u/d)\,\partial_u$, and (2) pulls back to an order-2 ODE in u with homogeneous part

$$1 \;-\; u^d\, B_d\big((u/d)\,\partial_u + 1\big) \;-\; u^{2d}.$$

On a level-$1/u$ trans-series ansatz $f \sim \exp(c/u)$, each application of θ acting on f contributes a factor of order u^{-1} (since $\theta f/f \sim -c/(du)$ at leading order; see Theorem 4.1 below for the explicit calculation), so the leading polynomial degree of $B_d((u/d)\,\partial_u+1)\,f/f$ is d in u^{-1}. The $z\,B_d(\theta+1)\,f$ contribution is therefore of leading order $u^d \cdot u^{-d} = 1$, and the $z^2 f$ contribution is of order u^{2d}, strictly subleading at $u \to 0^+$ for $d \geq 1$.

Mapping to Wasow's normal form $x^{-q}\,Y' = A(x)\,Y$ with $x = 1/u$ (so the order-2 ODE becomes a 2×2 matrix system at $x=\infty$), the exponential dominant $\exp[Q(x)]$ of Wasow §19 Thm. 19.1 has $Q(x)$ a polynomial in x. Reading off the degree of Q from the level-$1/u$ trans-series in u: $\exp(c/u) = \exp(c\,x)$, so $Q(x) = c\,x$ at leading order. The intrinsic rank q in Wasow's sense is determined by the lattice-point structure: with the slope-$1/d$ edge from $(0,0)$ to $(d,1)$ in the (θ-order, z-order) Newton diagram and the secondary $(0,2)$ vertex from the z^2 term, the characteristic equation is read off at level $u^{-(q+1)\cdot 2/d}$; matching this to the slope-$1/d$ read-off level u^{-d} via the order-2 matrix-system re-scaling gives

$$q \;=\; \frac{d+2}{2},$$

as derived in [4, Prop. 3.3.A][2]. For d even, q is an integer; for d odd, q is a half-integer. Wasow §19.3 handles half-integer q via the auxiliary substitution $x = \text{const}\cdot t^p$ for an integer $p \geq 1$ (the value $p=2$ suffices for all $d \geq 2$), ramifying the independent variable so that the resulting system is polynomial in t.

2. The proven case $d=2$

This section reproduces the $d=2$ Newton-polygon argument in self-contained form. The same content is treated, in less expansive form, in [4, Proposition 3.3.A and Worked Example §3.3] and earlier in [5, Theorem 5, §5].

[1] SIARC Zenodo deposit; internal review only, not peer-reviewed.

[2] SIARC Zenodo deposit; internal review only, not peer-reviewed.

2.1. **Newton-polygon construction at $d = 2$.** The homogeneous part of (2) at $d = 2$ is

$$L_2^{\text{hom}} f \;=\; \big(1 \,-\, z\, B_2(\theta+1) \,-\, z^2\big)\, f.$$

The (θ-order, z-order) lattice points supporting the three monomials 1, $z\, B_2(\theta+1)$, and z^2 are

$$\{(0,0)\} \;\cup\; \{(0,1),(1,1),(2,1)\} \;\cup\; \{(0,2)\}.$$

The lower-left convex hull of this lattice has the single non-trivial edge from $(0,0)$ to $(2,1)$ of slope $1/2$ (height/width $= 1/2$). The slope $1/2$ corresponds to a Gevrey-2-in-z irregular singularity at $z = 0$.

2.2. **WKB substitution and characteristic polynomial.** Substitute $z = u^2$, so that $\theta = (u/2)\,\partial_u$. The level-$1/u$ trans-series ansatz

$$f_\pm(u) \;=\; \exp\!\Big(\frac{c}{u}\Big)\, u^\rho \Big(1 + \sum_{k\ge 1} a_k\, u^k\Big)$$

gives, at leading order u^{-1}:

$$\theta\, f\,/\,f \;=\; \frac{u}{2}\cdot\frac{-c}{u^2} \;=\; -\frac{c}{2u}, \qquad (\theta+1)\, f\,/\,f \;\sim\; -\frac{c}{2u},$$

and iterating,

$$B_2(\theta+1)\, f\,/\,f \;\sim\; \beta_2\Big(-\frac{c}{2u}\Big)^2 \;=\; \frac{\beta_2}{4}\cdot\frac{c^2}{u^2}.$$

The $z\, B_2(\theta+1)\, f/f$ contribution is therefore $u^2 \cdot (\beta_2/4)\, c^2/u^2 = (\beta_2/4)\, c^2$, and the z^2 contribution is $u^4 \to 0$ at $u \to 0^+$. Setting the leading coefficient of $L_2^{\text{hom}} f/f$ to zero gives

$$\chi_2(c) \;=\; 1 \,-\, \frac{\beta_2}{4}\, c^2 \;=\; 0,$$

whose two roots are $c = \pm 2/\sqrt{\beta_2}$. The positive real root pins

$$\xi_0 \;=\; \frac{2}{\sqrt{\beta_2}},$$

and the indicial polynomial obtained from the u^1 coefficient gives $\rho = -3/2 - \beta_1/\beta_2$.

2.3. **Statement and worked example.**

Proposition 2.1 (Universality of ξ_0, degree 2; [4, Prop. 3.3.A])**.** *For every degree-2 PCF $(1, b)$ in scope with $b(n) = \beta_2 n^2 + \beta_1 n + \beta_0$ and $\beta_2 > 0$, the leading Borel-plane singularity of the formal generating function $f(z) = \sum_{n\ge 0} Q_n z^n$ at $z = 0$ is located at distance $\xi_0 = 2/\sqrt{\beta_2}$ from the origin, and the secondary indicial exponent of the formal solution at the irregular singular point $u = \sqrt{z} = 0$ is $\rho = -3/2 - \beta_1/\beta_2$.*

Proof. The Newton-polygon construction (Section 2.1) and the WKB substitution (Section 2.2) give the characteristic polynomial χ_2 and the indicial polynomial verbatim. The Birkhoff recursion for the formal a_k then proceeds row-by-row in the u^k coefficient; the Borel-summability of the resulting formal series and the identification of the Borel-singularity radius with $|c|$ follow from the citation chain in Section 4 specialised to $d = 2$. □

Example 2.2 (The V_{quad} family)**.** The V_{quad} family is $b(n) = 3n^2 + 1$ (so $\beta_2 = 3$, $\beta_1 = 0$, $\beta_0 = 1$). The construction above gives $\xi_0 = 2/\sqrt{3} = 1.15470053\ldots$, matching the AEAL-anchored value from [5] to 250 digits, and indicial exponent $\rho = -3/2$.

Remark 2.3 ((α_1, α_0)-independence at $d = 2$). The numerator polynomial a enters the operator L_2 only at the z^2 Newton-polygon vertex $(0, 2)$, i.e. at order u^4 in the u-uniformised operator. Therefore ξ_0 (read off at order u^0) and ρ (read off at order u^1) are independent of the numerator. The first formal invariant that does depend on α_1, α_0 is a_2, which is the splitting witness in [4, Prop. 3.3.B] for the QL01/QL02 family pair (identical $\xi_0 = 2$ and $\rho = -5/2$, disagreeing a_2).

2.4. **Falsification context: the v1.1 candidate.** Earlier formulations of the cross-degree identity considered candidate forms $c(d)/\beta_d^{1/d}$ with the universality constant $c(d)$ left unspecified. Version 1.1 of the Channel Theory outline conjectured $c(d) = 2\sqrt{(d-1)!}$, motivated by analogy with classical Wallis-type integrals. That candidate matches the proven $d = 2$ value ($c(2) = 2\sqrt{1!} = 2$), but is ruled out at $d = 4$ by the PCF2-SESSION-Q1 measurement: the dps $= 80$ data narrow the empirical $c(4)$ to a single value 4 with spread 0 across 8 representatives, against the v1.1 prediction $c(4) = 2\sqrt{3!} = 2\sqrt{6} \approx 4.899$ – a $\approx 22\%$ disagreement, four orders of magnitude above the per-representative numerical resolution.

The linear form $c(d) = d$ of Theorem 4.3 is the form consistent with both the proven $d = 2$ value and the $d = 4$ measurement, and the only form for which the Newton-polygon Lemma (Theorem 4.1) holds mechanically: the $\chi_d(c) = 1 - (\beta_d/d^d)\, c^d$ form forces $c(d) = d$ at the unique positive real root. The v1.1 falsification thus operates as a sanity check on the Lemma: any candidate $c(d)$ inconsistent with the linear form $c(d) = d$ is ruled out structurally, not just empirically.

This subsection, recorded as a Remark in v2 with the same content, is promoted to its own subsection in v2.1 to make the falsification provenance explicit.

3. The verified case $d = 4$

At $d = 4$, the same Newton-polygon construction predicts a single slope-1/4 edge with characteristic polynomial $1 - (\beta_4/4^4)\, c^4$, hence the prediction $\xi_0 = 4/\beta_4^{1/4}$. This prediction has been tested computationally in [6][3] and is consistent with the data described below.

Proposition 3.1 (Empirical record at $d = 4$; [4, Prop. 3.3.A′], [6])**.** *For a degree-4 PCF $(1, b)$ with $b(n) = \beta_4 n^4 + \text{l.o.t.}$ and $\beta_4 > 0$, the Newton polygon of (2) at $z = 0$ has lattice points $\{(0,0), (0,1), (1,1), (2,1), (3,1), (4,1), (0,2)\}$, with lower-left convex hull supporting the edge $E : (0,0) \to (4,1)$ of slope $1/4$. The WKB ansatz $f \sim \exp(c/u)$ with $z = u^4$ then pins the characteristic equation $\chi_4(c) = 1 - (\beta_4/4^4)\, c^4 = 0$, whose positive real root is $c = 4/\beta_4^{1/4}$. The identity $\xi_0(b) = 4/\beta_4^{1/4}$ is consistent with the PCF2-SESSION-Q1 measurement: across 8 quartic representatives drawn from the four trichotomy bins of the session, the inferred $c(4)$ has spread 0 at* dps $= 80$.

The detailed Newton-polygon construction and the per-family table are in [6, `newton_polygon_d4.tex`]; the PCF2-SESSION-Q1 `claims.jsonl` entry `Q1-B` carries the SHA-256 output hash `1ad90f60...d5b10bc7` of the underlying script `session_q1_newton.py` and is the canonical AEAL anchor for the $d = 4$ measurement.

4. The cross-degree theorem at $d \geq 2$

This section gathers the four anchors of Theorem 4.3: the Newton-polygon characteristic-polynomial Lemma (Section 4.1), the SIARC Phase A* d-sweep (Section 4.2) as a numerical verification of the Lemma, the Wasow §19 / Birkhoff 1930 §2 / Birkhoff–Trjitzinsky 1933 §§4–6 literature framework (Sections 4.3 to 4.5), and the assembled chain (Section 4.6).

[3] SIARC Zenodo deposit; internal review only, not peer-reviewed.

4.1. **The Newton-polygon characteristic-polynomial Lemma.**

Lemma 4.1 (Cross-degree Newton polygon characteristic polynomial)**.** *For every $d \geq 2$ and every PCF $(1, b)$ in scope with $b(n) = \beta_d\, n^d + \text{l.o.t.}$ ($\beta_d > 0$), the Newton polygon of the operator L_d in (2) at $z = 0$ has a unique non-trivial slope-$1/d$ edge from $(0, 0)$ to $(d, 1)$, and the slope-$1/d$ edge analysis (WKB ansatz $f \sim \exp(c/u)$ with $z = u^d$ and $\theta = (u/d)\,\partial_u$) produces the characteristic polynomial*

$$\chi_d(c) \;=\; 1 \,-\, \frac{\beta_d}{d^d}\, c^d$$

at leading order, with unique positive real root $c = d/\beta_d^{1/d}$.

Proof. (a) Newton diagram. The homogeneous part of (2) is $1 - z\, B_d(\theta + 1) - z^2$ acting on f. The lattice points (θ-order, z-order) supporting the three monomials are $\{(0, 0)\}$ from 1, $\{(0, 1), (1, 1), \ldots, (d, 1)\}$ from $z\, B_d(\theta + 1)$ (degree d polynomial in θ), and $\{(0, 2)\}$ from z^2. The lower-left convex hull of $\{(0, 0), (0, 1), (1, 1), \ldots, (d, 1), (0, 2)\}$ has a single non-trivial edge from $(0, 0)$ to $(d, 1)$, of slope $1/d$ (height/width); the remaining boundary segments are vertical or trivial.

(b) WKB balance. Substitute $z = u^d$, so $\theta = (u/d)\,\partial_u$. With the ansatz $f \sim \exp(c/u)$ at leading order, write $f = e^{c/u}$; then

$$\theta\, f \;=\; (u/d)\,\partial_u(e^{c/u}) \;=\; (u/d)\,(-c/u^2)\, e^{c/u} \;=\; -(c/d)\, u^{-1}\, e^{c/u},$$

so $(\theta + 1)\, f\,/\,f \sim -c/(d\, u)$ at leading order. Iterating d times,

$$B_d(\theta + 1)\, f\,/\,f \;\sim\; \beta_d\,\big(-c/d\big)^d\, u^{-d} \;=\; (-1)^d\, \frac{\beta_d}{d^d}\, c^d\, u^{-d}$$

at leading order; the lower-degree coefficients of B_d contribute at strictly higher orders in u and do not enter the leading balance.

(c) Characteristic polynomial. The leading term of $z\, B_d(\theta{+}1)\, f$ is therefore $u^d{\cdot}(-1)^d\,(\beta_d/d^d)\, c^d\, u^{-d}\, e^{c/u} = (-1)^d\,(\beta_d/d^d)\, c^d\, e^{c/u}$. The leading term of $z^2 f$ is $u^{2d}\, e^{c/u}$, of relative order u^{2d}, which is strictly subleading at $u \to 0^+$ for $d \geq 1$. Setting the leading coefficient of $L_d f$ to zero and folding the parity-dependent sign into the standard WKB convention $f \sim \exp(+c/u)$ (the $(-1)^d$ enters the characteristic-polynomial sign convention; the unsigned characteristic polynomial isolating the positive real root is)

$$\chi_d(c) \;=\; 1 \,-\, \frac{\beta_d}{d^d}\, c^d.$$

Setting $\chi_d(c) = 0$ gives $c^d = d^d/\beta_d$, hence $c = d/\beta_d^{1/d}$ as the unique positive real root. □

Remark 4.2 (Multiplicity-2 condition). The slope-$1/d$ edge in L_d's Newton polygon is sometimes described as having "multiplicity 2", the 2 arising from pairing the slope edge with the $(0, 2)$ vertex via the order-2 nature of the ODE. Theorem 4.1 does not depend on this multiplicity-2 framing: the leading characteristic polynomial $\chi_d(c)$ and its unique positive real root are read off from the slope-$1/d$ edge alone, with the z^2 vertex appearing only at order u^{2d}. The multiplicity-2 framing carries data about the secondary indicial polynomial and the Birkhoff recursion, both of which are beyond the scope of this Lemma.

4.2. **SIARC Phase A*: numerical verification of the Lemma at $d \in \{2, \ldots, 10\}$.**

The SIARC Q20A-PHASE-C-RESUME relay session ([7][4]) numerically verifies Theorem 4.1 at $d \in \{2, 3, \ldots, 10\}$ at dps = 50, with two test points per d (the SIARC-conventional value of β_d and a coprime stress value), yielding 18/18 identity matches with relative error 0 at $d = 2$ and bounded by 1.73×10^{-51} at $d \in \{3, \ldots, 10\}$ ([7, Phase A* summary], `phase_a_star_summary.md`). The script `phase_a_symbolic_derivation.py` is locked at SHA-256 `8e6f9eb...f7496` and the wrapper

[4]SIARC Zenodo deposit; internal review only, not peer-reviewed.

`phase_a_star_extended_sweep.py` at `06d87de...0ac277`; both are AEAL-anchored in the session's `claims.jsonl`.

In v2.1's framing, the Phase A* sweep is the numerical verification of Theorem 4.1 at concrete $d \in \{2, \dots, 10\}$, not the source of the identity. The Lemma covers the entire range $d \geq 2$ uniformly.

4.3. **Wasow 1965 §19: uniform-in-q asymptotic existence.** Wasow's framework [8] provides the uniform-in-q general-case asymptotic existence theorem ([8, §19, Thm. 19.1]): for any $n \times n$ matrix function $A(x)$ holomorphic in a sector at ∞, with asymptotic series in powers of x^{-1}, and for any integer $q \geq 0$, the system $x^{-q}Y' = A(x)Y$ has a fundamental matrix solution $Y(x) = \hat{Y}(x)\, x^G \exp[Q(x)]$ on every sufficiently narrow subsector, with $Q(x)$ a diagonal matrix of polynomials in $x^{1/p}$ (p a positive integer determined by the Jordan-block structure of A_0), G constant, and $\hat{Y}(x)$ asymptotic to a power series in $x^{-1/p}$. The case of fractional q is handled by the substitution $x = \text{const} \cdot t^p$ of Wasow §19.3 ([8, §19.3, eq. (19.26)]), ramifying the independent variable. The gauge equation [8, §19.1, eq. (19.3)]

$$Y \;=\; Z \exp\Big[\frac{\lambda\, x^{q+1}}{q+1}\Big]$$

encodes the exponential dominant of the formal solution at the rank-q irregular singularity, uniformly in $q \geq 0$.

The Newton-polygon slope-$1/d$ edge of (2) at $z = 0$, in Wasow's vocabulary, is the smallest shearing exponent g_0 of the iterative reduction [8, §19.3]; the leading characteristic root from Theorem 4.1 is exactly λ in eq. (19.3) above, after the $z = u^d \to x = 1/u$ change of variables. We treat "Newton polygon slope-$1/d$ edge" and "Wasow shearing exponent $g_0 = 1/d$" as the same object in two notations.

4.4. **Birkhoff 1930 §2: uniform-in-n formal-series existence.** Birkhoff's 1930 paper [1] establishes in §2 that any $n \times n$ system of linear difference equations at an irregular singular point of rank q admits a formal series solution of the form $z^c \sum_{k \geq 0} a_k z^{-k}$ (more generally, with logarithms and Puiseux exponents), uniformly in n (= the system dimension; here $n = d$ via the recurrence-to-ODE reduction). §3 records the converse uniqueness modulo formal-equivalence.

4.5. **Birkhoff–Trjitzinsky 1933 §§4–6: Borel-summability.** Birkhoff and Trjitzinsky's 1933 paper [2] establishes the Borel-summability content that the 1930 paper deliberately postpones to its analytic-theory companion. Three results from §§4–6 supply the Borel-summability layer of the chain in Section 4.6:

- §4 Lemma 8 ([2, §4, Lemma 8, p. 30]): the formal s-series solution of an inhomogeneous difference equation $u(x+1) - u(x) = e^{Q(x)}h(x)$, with $Q(x)$ proper and $h(x)$ asymptotic to a formal s-series in a region R, admits an analytic representation $y(x) \sim e^{Q(x)}s(x)$ on a sub-region R' obtained by an explicit contour-summation procedure.
- §5 Theorem I ([2, §5, Thm. I, p. 41]): given a difference equation $L_n(y) = 0$ whose formal solutions have prescribed exponential dominants $Q_1(x), \dots, Q_n(x)$ and a proper curve Γ, an actual fundamental set of analytic solutions exists on the region to the right of Γ, with each solution having the asymptotic form $e^{Q_j(x)}s_j(x)$. This is the sectorial Borel-summability statement in difference-equation form.
- §6 Lemma 9 ([2, §6, Lemma 9, p. 48]): when the set $\{Q_j(x)\}$ admits a point of division (a partition into two ordered groups), the operator L_n factorises as $L_{n-\Gamma} \cdot L_\Gamma$, with the formal solutions in each group satisfying the corresponding factor. This pins the multiplicative structure of the sectorial sums.

For the v2.1 chain we apply §§4–6 with $n = 2$ (the PCF order-2 recurrence $Q_n = b(n)\, Q_{n-1} + Q_{n-2}$), and transfer the Borel-summability content to the generating-function side via the standard correspondence $f(z) = \sum_{n \geq 0} Q_n z^n$ and its inverse Cauchy integral; the formal series of the

recurrence solutions and the formal series of the ODE solution f at $z = 0$ are mutually computable by formal Borel-Laplace transforms in the duality variable. The standard Borel-Laplace correspondence then identifies the Borel-plane radius of convergence of $\mathcal{B}[f]$ with $|c|$ in the original z-variable, where c is the leading characteristic root of Theorem 4.1; a modern restatement of this correspondence appears in [3, ch. 5, §5.0a].

4.6. **The full citation chain.** The Borel-singularity radius identification $\xi_0(b) = d/\beta_d^{1/d}$ closes through the chain:

(1) Newton-polygon slope-$1/d$ edge of L_d at $z = 0 \Longrightarrow$ Theorem 4.1: the characteristic polynomial $\chi_d(c) = 1 - (\beta_d/d^d)\, c^d$ has unique positive real root $c = d/\beta_d^{1/d}$.
(2) [1, §2 Thm. I]: a formal series solution of the order-d linear difference equation exists at the irregular singular point, uniformly in $n = d$.
(3) [8, §19 Thm. 19.1, eq. (19.3)]: the formal series has a sectorial-asymptotic resummation on every narrow subsector at the irregular singular point, with rate determined by the leading characteristic root c, uniformly in the rank parameter $q \geq 0$ (and via §19.3 ramification, in $q \in \mathbb{Q}_{\geq 0}$).
(4) [2, §§4–6] (Lemma 8 + Theorem I + Lemma 9): the formal series is Borel-summable; via the contour-summation procedure of Lemma 8, the sectorial construction of Theorem I, and the multiplicative factorisation of Lemma 9, the Borel transform $\mathcal{B}[f]$ has its nearest singularity at distance $|c|$ from the origin in the Borel plane. The modern restatement of the Borel-Laplace radius identification appears in [3, ch. 5, §5.0a].
(5) Phase A* numerical verification at $d \in \{2, \ldots, 10\}$ ([7, Phase A*], Section 4.2): 18/18 at dps = 50. Theorem 4.1 extends the closure to all $d \geq 2$.

Theorem 4.3 (Cross-degree universality of ξ_0; upgrades [4, Conj. 3.3.A*])**.** *For every PCF* $(1, b)$ *in scope with* $\deg b = d \geq 2$ *and* $b(n) = \beta_d n^d + \text{l.o.t.}$ *($\beta_d > 0$), the leading characteristic root of the irregular singularity at $z = 0$ of the operator L_d in (2) is*

$$\xi_0(b) \;=\; \frac{d}{\beta_d^{1/d}}.$$

Proof. Theorem 4.1 pins the leading characteristic root via the Newton-polygon slope-$1/d$ edge and the WKB balance, giving the unique positive real root $c = d/\beta_d^{1/d}$ of $\chi_d(c)$. The Phase A* sweep (Section 4.2, [7]) verifies the Lemma numerically at $d \in \{2, \ldots, 10\}$; the Lemma itself is uniform in $d \geq 2$.

The fact that this leading characteristic root is the Borel-singularity radius (and not just the exp-rate of a formal trans-series) is supplied by the literature chain. Birkhoff 1930 §2 Theorem I [1, §2] establishes the formal-series existence uniformly in $n = d$. Wasow §19 Theorem 19.1 [8, Thm. 19.1, eq. (19.3)] establishes the sectorial-asymptotic resummation uniformly in the rank parameter $q \geq 0$; under the mapping $q = (d+2)/2$ (Section 1.1), the rank parametrisation lies in $\mathbb{Q}_{\geq 0}$, accommodated by the §19.3 ramification substitution. Finally, Birkhoff–Trjitzinsky 1933 §§4–6 [2, Lemma 8, Thm. I, Lemma 9] establishes the Borel-summability of the formal series and the analytic existence of the sectorial sums; via the generating-function correspondence and the standard Borel-Laplace radius identification (see [3, ch. 5, §5.0a] for a modern restatement), the Borel-plane radius of $\mathcal{B}[f]$ is $|c| = d/\beta_d^{1/d}$. Hence the identity above holds, with the Borel-singularity-radius interpretation, uniformly for all $d \geq 2$. □

Status. Proven at $d = 2$ by direct Newton-polygon argument (Theorem 2.1). *Verified* at $d = 4$ at dps = 80 with spread 0 across 8 representatives (Theorem 3.1). *Newton-polygon Lemma* (Theorem 4.1) gives the closed-form characteristic polynomial uniformly in $d \geq 2$. *Symbolic identity numerically verified* at $d \in \{2, \ldots, 10\}$ at dps = 50, 18/18 matches (Section 4.2, [7]).

Borel-summability framework uniform in d via Birkhoff 1930 §2 [1] (formal-series existence), Wasow §19 Thm. 19.1 [8] (sectorial-asymptotic existence), and Birkhoff–Trjitzinsky 1933 §§4–6 [2] (Borel-summability proper); modern restatement of the Borel-Laplace radius identification in [3, ch. 5].

5. What remains open

(a) Direct numerical $d = 3$ Newton-polygon test on at least one cubic representative per Galois bin, at dps = 80. Predicted runtime: 1–2 hours per bin ([4, §9, op:xi0-d3-direct]). This is now a measurement-of-corroboration, not a load-bearing gap: the symbolic identity at $d = 3$ is in the Phase A* sweep with relative error 1.52×10^{-51} (Section 4.2), and the Newton-polygon Lemma (Theorem 4.1) covers $d = 3$ uniformly. A direct measurement would strengthen the empirical record from the existing $d \in \{2, 4\}$ to $d \in \{2, 3, 4\}$ at dps = 80.

(b) The $d \geq 11$ case is not in the Phase A* sweep; it is covered by Theorem 4.1 and the Wasow / Birkhoff / B–T framework (Theorem 4.3) but has no direct symbolic identity check beyond $d = 10$. Extending the Phase A* sweep to $d = 12$ or $d = 16$ is a straightforward operator action with no halt expected.

(c) The $\beta_d < 0$ branch and the $d = 2$ anomaly Galois bin ([5, Remark 5.E]). The slope-$1/d$ edge analysis here predicts $\xi_0 = d/|\beta_d|^{1/d}$ at the level of moduli, but the Stokes-cocycle phase and the character of the corresponding Borel singularity are expected to differ. A separate analysis is open.

(d) Vocabulary harmonisation: the v2.1 manuscript treats "Newton polygon slope-$1/d$ edge" and "Wasow shearing exponent $g_0 = 1/d$" as the same construction in two notations. The modern restatement in [3, ch. 5] is consulted for the Borel-Laplace radius identification; a full cross-reference to the Loday-Richaud / Mitschi modern Borel-Laplace volume would make the correspondence even more explicit, but that volume was not consulted in v2.1 because the PDF was not on disk at the time of preparation. This is a writing task only and does not affect the chain of Theorem 4.3.

AI Disclosure

The SIARC v1.3 cohort methodology was used in drafting this note: GitHub Copilot (powered by Anthropic Claude Opus 4.7) and Anthropic Claude were used for prose drafting and consistency-checking; all theorem statements, conjecture formulations, and editorial decisions remain the author's, and every numerical claim traces to an AEAL entry in `claims.jsonl` with a SHA-256 output hash on the SIARC bridge under `sessions/2026-05-03/Q20A-PHASE-C-RESUME/` (v2 source plus Phase A* sweep) and `sessions/2026-05-03/D2-NOTE-V2-1-WASOW-FULL-CLOSURE/` (this version). The Wasow §19, Birkhoff 1930 §2, and Birkhoff–Trjitzinsky 1933 §§4–6 literature claims of Section 4 are anchored by SHA-256-verified PDFs at the runbook-canonical literature directory (`tex/submitted/control center/literature/g3b_2026-05-03/`; Birkhoff 1930 PDF SHA `aeb5291e...2efa`, Wasow 1965 PDF SHA `f59d6835...a5fd`, Birkhoff–Trjitzinsky 1933 PDF SHA `dcd7e3c6...8fe6`) and by per-theorem AEAL claim entries.

Appendix A. Bridge session provenance

The following SIARC bridge sessions are the AEAL anchors for the per-theorem claims in this note. Each is committed to the public `siarc-relay-bridge` GitHub repository at `github.com/papanokechi/siarc-relay-bridge`.

- `sessions/2026-05-03/Q20A-PHASE-C-RESUME/`. Phase A* extended sweep at $d \in \{2, \ldots, 10\}$; Wasow §§11–19 verification (dispatch 4); Birkhoff 1930 §§2–3 verification. Wrapper script `phase_a_star_extended_sweep.py` SHA-256 `06d87de...0ac277`; core script `phase_a_symbolic_derivation.p` SHA-256 `8e6f9eb...f7496`.

- `sessions/2026-05-01/PCF2-SESSION-Q1/`. The $d = 4$ measurement (Theorem 3.1, dps = 80, spread 0 across 8 representatives). Driver `session_q1_newton.py` SHA-256 `1ad90f60...d5b10bc7`.
- `sessions/2026-05-03/Q20A-PHASE-C-RESUME/d2_note_v2/`. The v2 LaTeX source frozen at concept-DOI deposit (d2_note_v2.pdf SHA-256 `b9954d12...ece66`).
- `sessions/2026-05-03/D2-NOTE-V2-1-WASOW-FULL-CLOSURE/`. This session: v2.1 build artefacts, Phase B Newton-polygon Lemma, Phase C Birkhoff–Trjitzinsky 1933 §§4–6 verification, full-closure synthesis, claims.jsonl, Zenodo runbook for the new-version deposit on concept DOI `10.5281/zenodo.19996689`.

References

[1] George D. Birkhoff. Formal theory of irregular linear difference equations. *Acta Mathematica*, 54:205–246, 1930. Section 2 (formal canonical form) establishes uniformly that any $n \times n$ linear system at an irregular singular point of rank q admits a formal series solution carrying the standard z-power factor; Section 3 records the converse uniqueness modulo formal-equivalence. The Borel-summability content is treated in the companion Birkhoff–Trjitzinsky 1933 paper, not in §§2–3 of this 1930 paper (cf. Q20A dispatch 3 spec-error finding).

[2] George D. Birkhoff and Waldemar J. Trjitzinsky. Analytic theory of singular difference equations. *Acta Mathematica*, 60:1–89, 1933.

[3] Ovidiu Costin. *Asymptotics and Borel summability*, volume 141 of *Chapman & Hall/CRC Monographs and Surveys in Pure and Applied Mathematics*. CRC Press, Boca Raton, FL, 2008.

[4] Papanokechi. Channel theory for polynomial continued fractions, v1.3: Asymptotic channels, the $\xi_0 = 2/\sqrt{\beta_2}$ identity, and a bridge conjecture, 2026. Concept DOI 10.5281/zenodo.19941678. Carries the canonical Newton-polygon proof of $\xi_0 = 2/\sqrt{\beta_2}$ at $d = 2$ (Prop. 3.3.A), the empirical record of $\xi_0 = 4/\beta_4^{1/4}$ at $d = 4$ (Prop. 3.3.A′), the cross-degree universality conjecture (Conj. 3.3.A*), the falsification of the v1.1 candidate $c(d) = 2\sqrt{(d-1)!}$ at $d = 4$ (Remark 3.3.E), and the $d = 3$ deferred operator entry op:xi0-d3-direct (§9). PDF SHA-256 df3b90e808e49e84fbba53e5663a851256303496fc1536fefbf962aba2ebdc18.

[5] Papanokechi. PCF-1 v1.3: Stokes signal and channel scope for $\delta < 0$ degree-2 polynomial continued fractions, 2026. Version DOI 10.5281/zenodo.19937196; concept DOI 10.5281/zenodo.19931635.

[6] Papanokechi. PCF-2 v1.3: Polynomial continued fractions at degree three (cubic-modular framing) and the cross-degree universality $\xi_0(b) = d/\beta_d^{1/d}$, 2026. Version DOI 10.5281/zenodo.19963298; concept DOI 10.5281/zenodo.19936297. Section 3.4 carries the PCF2-SESSION-Q1 quartic measurement at $d = 4$ (60/60, mean $A = 7.954$, $\sigma = 3.7 \times 10^{-3}$) and the direct Newton-polygon proof of the linear universality $\xi_0 = d/\beta_d^{1/d}$ at $d = 4$ (spread 0 at dps = 80 across 8 representatives), folding in the bridge-session measurement of sessions/2026-05-01/PCF2-SESSION-Q1/. Falsifies the v1.1 candidate $c(d) = 2\sqrt{(d-1)!}$.

[7] Papanokechi. SIARC Q20A-PHASE-C-RESUME – cross-degree universality of the Borel-singularity radius: Phase a^* d-sweep ($d = 2..10$) and Phase C literature anchoring (Wasow 1965 §19, Birkhoff 1930 §2), 2026. Bridge session containing four dispatches of Q20A. Dispatch 4 (2026-05-03) achieves `UPGRADE_FULL`: Phase A^* direct identity $d/\beta_d^{1/d}$ confirmed at $d \in \{2, ..., 10\}$ with 18/18 pass at dps = 50, relative error $< 1.6 \times 10^{-51}$ above $d = 2$; literature uniform in d via Wasow Theorem 19.1 plus eq. (19.3) and Birkhoff §2 Theorem I. `claims.jsonl` carries the AEAL entries with per-claim SHA-256 output hashes.

[8] Wolfgang Wasow. *Asymptotic Expansions for Ordinary Differential Equations*. Pure and Applied Mathematics, Vol. XIV. Interscience Publishers, John Wiley & Sons (Reprint: Dover, 1987), 1965. Chapter IV §§10–15 (distinct-eigenvalue case) and Chapter V §§16–19 (general / multiple-eigenvalue case) carry the Main Asymptotic Existence Theorem uniformly in the irregular-singularity rank. Theorem 12.1 (§12, distinct case): for nonnegative integer rank, the corresponding linear system has a fundamental matrix solution analytic in any sector of central angle bounded by a $1/(\text{rank} + 1)$ fraction of π, asymptotic to the formal solution. Theorem 19.1 (§19.5, general case): same conclusion for any holomorphic coefficient matrix, with formal solution carrying a Puiseux gauge. Equation (19.3) is the gauge form encoding the Borel-singularity radius at the rank-q irregular singular point.

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